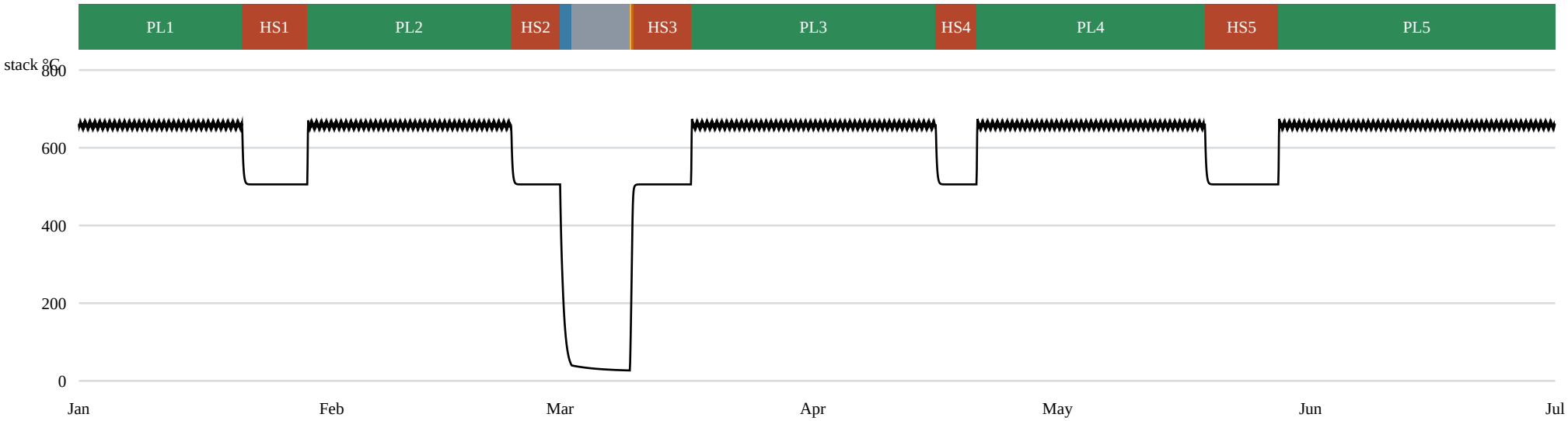
Six months of plant data from the twin

Answer: the twin app, unchanged, ran from 1 January to 1 July 2027, one row per minute. It had 5 part loads, 5 hot standbys and 1 cold shutdown, with no trips. The plant was available **95.00 %** of the time.

What the plant did, month by month



Coloured strip = plant stage every minute (PL = part load, HS = hot standby). The line is the stack temperature, one point per hour. The dip in March is the cold shutdown: the stack cools to 40 °C, sits for a week, then a cold start brings it back.

How the 95 % comes out

Available = part load + hot standby + fuel switch, divided by the whole period (same as the plant's availability check). Only the cold shutdown is not available. Its stay was set so that it uses exactly 5 % (13,032 minutes).

Stage	Minutes	Days	Share	Counts as
Part load	197,169	136.9	75.65 %	available
Hot standby	50,365	35.0	19.32 %	available
Fuel switch	76	0.1	0.03 %	available
Air flow cooling	2,035	1.4	0.78 %	not available
Shutdown	10,262	7.1	3.94 %	not available
Heat up air	272	0.2	0.10 %	not available
Heat up N2+H2	461	0.3	0.18 %	not available

Period	Starts (day)	Lasts (days)
PL1	0.0	20.0
HS1	20.0	8.0
PL2	28.0	25.0
HS2	53.0	6.0
HS3	68.0	7.0
PL3	75.1	30.0
HS4	105.1	5.0
PL4	110.1	28.0
HS5	138.1	9.0
PL5	147.1	33.9

If you remember one thing: 181 days of twin data, 5 part loads, 5 hot standbys, 1 cold shutdown, and 95.00 % available.

How the six months were made

A small script opens the twin app without a screen and clicks the same buttons a person would, at set times. The model, stage rules and sensor tables are the ones in the app today (version 2.0).

Step	When	What happens
1. Warm-up	not recorded	Start plant → heat up → hot standby → load, settle 2 days
2. PL1 → HS1 → PL2 → HS2	Jan – Feb	load, then current to 0: 10-min unload to hot standby
3. Cold shutdown	1 – 9 Mar	Shut down: air 9000, air off below 40 °C, stay, Start plant
4. HS3 → PL3 → HS4 → PL4 → HS5 → PL5	Mar – Jun	same buttons; PL5 runs to 1 July

The cold start takes 735 minutes (about 12.2 h), from 25 °C to hot standby.

Why the record starts in part load

In the twin, fuel switch starts only from hot standby, and part load always unloads back to hot standby. So every part load sits between two hot standbys. If the record began with the cold start, 5 part loads would bring 6 hot standbys. The first start-up and loading happen before minute 0 and are not saved.

What you get

04_synthetic_data/twin_6month/: sofc_twin_6month.csv.gz (260,640 rows × 117 columns: timestamp, minute, plant_stage + the 114 sensors of Save data), events.csv (every stage change), summary.json. Re-make for any length and availability: node 04_synthetic_data/twin_dataset.js --days 181 --availability 95 (about 40 s per 6 months; it stops with a message when the target cannot be met).

What to keep in mind

• The run is the same every time, with no random noise. Readings are flat during long holds: only 28 of 117 columns move during a part-load hold.
• Part load is the plant's median setting: 6.4 A per string, 14.6 kW in total, the same in all 5 runs.
• Part load lasts 20–34 days here. The real plant never held part load longer than 44 minutes. The twin's stack keeper holds it at about 660 °C; the safety reading peaks at 696.5 °C, below the 720 °C trip.
• After a week with the air off the stack is still at 27 °C, so the restart heats up from 27 °C. Availability comes out at 95.001 %.

If you remember one thing: the data comes from the app's own model and buttons. The only tuned number is how long the shutdown lasts, and that sets the 95 %.